

**SARGATSKOE, Russian Federation**WMO: **285980**Lat: **55.600N**Lon: **73.483E**Elev: **264**StdP: **14.56**Time Zone: **6.00 (E06)**Period: **90-05**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -31.9      | -25.8 | -36.7                         | 0.7 | -31.9 | -30.5 | 1.0 | -25.4 | 23.2                  | 24.5 | 21.0 | 23.4 | 3.4                   | 210  | 0.639 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 17.9                   | 86.4            | 66.3 | 83.1 | 65.5 | 79.8 | 64.1 | 69.4                | 81.6 | 67.6 | 78.6 | 66.0 | 76.4 | 8.6                  | 170  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 65.4                            | 94.7 | 73.6 | 63.7 | 89.0 | 72.1 | 62.0 | 83.9 | 70.5 | 33.7          | 81.6 | 32.2 | 78.8 | 30.9 | 76.5 | 75.9              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |    |    |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|----|----|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |    |    |
| (4) 22.4          | 19.7 | 17.3 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |    |    |
| (5)               |      |      | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |    |    |

**Monthly Climatic Design Conditions**

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 34.7   | 2.5   | 3.4   | 19.7  | 37.5  | 53.6  | 62.8  | 66.4  | 61.4  | 47.9  | 37.6  | 17.2  | 4.0   |
|                                                                                  | DBStd       | 25.80  | 14.47 | 14.63 | 11.43 | 11.17 | 10.23 | 8.55  | 6.41  | 7.40  | 7.46  | 9.14  | 14.24 | 12.80 |
|                                                                                  | HDD50       | 7136   | 1473  | 1306  | 940   | 394   | 81    | 9     | 0     | 5     | 126   | 391   | 985   | 1426  |
|                                                                                  | HDD65       | 11316  | 1938  | 1726  | 1405  | 825   | 373   | 141   | 61    | 156   | 515   | 850   | 1435  | 1891  |
|                                                                                  | CDD50       | 1542   | 0     | 0     | 0     | 19    | 193   | 393   | 508   | 360   | 62    | 7     | 0     | 0     |
|                                                                                  | CDD65       | 246    | 0     | 0     | 0     | 0     | 19    | 75    | 104   | 46    | 2     | 0     | 0     | 0     |
|                                                                                  | CDH74       | 2501   | 0     | 0     | 0     | 3     | 285   | 816   | 911   | 456   | 30    | 0     | 0     | 0     |
|                                                                                  | CDH80       | 678    | 0     | 0     | 0     | 1     | 68    | 233   | 249   | 123   | 4     | 0     | 0     | 0     |
| Wind                                                                             | WSAvg       | 7.4    | 7.7   | 7.7   | 8.1   | 8.2   | 8.0   | 6.6   | 6.1   | 5.7   | 6.9   | 8.4   | 7.9   | 7.9   |
| Precipitation                                                                    | PrecAvg     | 15.6   | 0.7   | 0.6   | 0.7   | 0.9   | 1.2   | 2.1   | 2.6   | 2.2   | 1.2   | 1.2   | 1.2   | 1.0   |
|                                                                                  | PrecMax     | 21.7   | 1.7   | 1.2   | 1.8   | 2.0   | 2.8   | 3.8   | 5.2   | 5.3   | 3.3   | 3.0   | 2.2   | 1.9   |
|                                                                                  | PrecMin     | 10.2   | 0.1   | 0.1   | 0.1   | 0.2   | 0.4   | 0.7   | 0.6   | 0.7   | 0.4   | 0.2   | 0.4   | 0.4   |
|                                                                                  | PrecStd     | 2.8    | 0.4   | 0.2   | 0.4   | 0.5   | 0.6   | 0.9   | 1.3   | 1.0   | 0.6   | 0.6   | 0.5   | 0.4   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 34.0  | 34.4  | 42.2  | 71.9  | 86.3  | 89.0  | 91.2  | 88.8  | 79.3  | 64.6  | 43.4  | 33.4  |
|                                                                                  |             | MCWB   | 32.4  | 32.0  | 36.7  | 55.6  | 63.2  | 66.1  | 69.8  | 67.4  | 59.1  | 49.9  | 40.9  | 32.9  |
|                                                                                  | 2%          | DB     | 30.1  | 31.6  | 38.8  | 65.1  | 80.8  | 85.7  | 86.1  | 83.1  | 70.3  | 59.1  | 40.3  | 30.4  |
|                                                                                  |             | MCWB   | 28.7  | 29.9  | 35.3  | 52.2  | 60.6  | 65.9  | 67.1  | 65.6  | 56.4  | 48.8  | 37.9  | 29.8  |
|                                                                                  | 5%          | DB     | 27.3  | 29.6  | 36.4  | 60.1  | 76.8  | 82.4  | 82.5  | 78.3  | 64.9  | 54.3  | 36.4  | 25.8  |
|                                                                                  |             | MCWB   | 26.2  | 28.4  | 34.0  | 49.4  | 58.5  | 65.2  | 66.6  | 63.9  | 53.9  | 46.6  | 34.5  | 24.8  |
|                                                                                  | 10%         | DB     | 23.4  | 26.3  | 34.6  | 55.1  | 72.3  | 78.8  | 79.2  | 74.8  | 60.0  | 50.8  | 34.0  | 21.9  |
|                                                                                  |             | MCWB   | 22.4  | 25.3  | 32.6  | 46.7  | 57.3  | 63.6  | 65.7  | 62.5  | 52.3  | 44.6  | 32.6  | 20.9  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 32.3  | 32.4  | 38.0  | 57.7  | 65.7  | 71.6  | 71.7  | 70.0  | 60.9  | 53.5  | 42.0  | 32.7  |
|                                                                                  |             | MCDB   | 33.4  | 33.9  | 40.9  | 67.7  | 83.4  | 84.8  | 83.1  | 82.5  | 74.7  | 61.0  | 43.1  | 33.0  |
|                                                                                  | 2%          | WB     | 28.8  | 30.2  | 35.7  | 53.9  | 61.9  | 68.6  | 70.1  | 67.3  | 57.5  | 50.0  | 38.0  | 29.9  |
|                                                                                  |             | MCDB   | 29.6  | 31.5  | 38.0  | 63.3  | 76.4  | 80.7  | 82.2  | 79.6  | 66.8  | 56.7  | 39.8  | 30.6  |
|                                                                                  | 5%          | WB     | 26.1  | 28.3  | 34.4  | 50.4  | 59.8  | 66.4  | 68.3  | 65.5  | 55.1  | 47.9  | 34.7  | 24.6  |
|                                                                                  |             | MCDB   | 27.2  | 29.6  | 36.2  | 59.0  | 74.2  | 78.2  | 79.0  | 75.6  | 62.3  | 53.2  | 36.2  | 25.5  |
|                                                                                  | 10%         | WB     | 22.3  | 25.3  | 32.8  | 46.9  | 57.8  | 64.5  | 66.7  | 63.8  | 52.9  | 45.4  | 32.5  | 20.9  |
|                                                                                  |             | MCDB   | 23.3  | 26.4  | 34.5  | 54.6  | 70.3  | 76.7  | 76.4  | 71.9  | 59.7  | 50.2  | 33.8  | 21.9  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 15.5  | 18.2  | 17.4  | 17.8  | 22.0  | 19.7  | 17.9  | 19.3  | 16.5  | 13.1  | 12.7  | 14.9  |
|                                                                                  |             | MCDBR  | 13.9  | 13.1  | 14.3  | 26.4  | 29.2  | 25.7  | 23.1  | 25.1  | 25.4  | 19.7  | 10.5  | 15.0  |
|                                                                                  |             | MCWBR  | 13.4  | 12.4  | 11.8  | 15.8  | 12.7  | 10.7  | 8.7   | 10.5  | 13.2  | 12.3  | 9.7   | 14.7  |
|                                                                                  | 5% WB       | MCDBR  | 14.2  | 13.0  | 12.5  | 25.0  | 26.7  | 23.2  | 19.9  | 20.9  | 21.5  | 18.1  | 10.4  | 14.2  |
|                                                                                  |             | MCWBR  | 13.8  | 12.4  | 10.9  | 16.0  | 12.9  | 11.4  | 8.9   | 10.3  | 12.8  | 12.4  | 10.0  | 14.1  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |        | 0.282 | 0.316 | 0.375 | 0.411 | 0.403 | 0.411 | 0.432 | 0.432 | 0.371 | 0.355 | 0.321 | 0.287 |
|                                                                                  | taud        |        | 2.038 | 2.012 | 1.922 | 2.095 | 2.226 | 2.282 | 2.220 | 2.187 | 2.322 | 2.220 | 2.085 | 1.991 |
|                                                                                  | Ebn at Noon |        | 189   | 228   | 241   | 254   | 266   | 265   | 255   | 246   | 247   | 215   | 170   | 150   |
|                                                                                  | Edh at Noon |        | 21    | 32    | 45    | 44    | 41    | 39    | 41    | 40    | 30    | 26    | 20    | 17    |
| All-Sky Solar<br>Radiation                                                       | RadAvg      |        | 285   | 625   | 1122  | 1465  | 1741  | 1871  | 1731  | 1376  | 949   | 490   | 240   | 184   |
|                                                                                  | RadStd      |        | 21    | 31    | 79    | 113   | 147   | 128   | 122   | 117   | 86    | 54    | 37    | 20    |

**Historical Trends**

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (13 neighbors) | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page